Federal Bank: Infrastructure Sizing and Volumetrics

Hosting: AWS, Asia Pacific (Mumbai) primary, Asia Pacific (Hyderabad) DR. Sizing basis: FY26-27 volumetrics shared by Federal Bank.

1. Volumetrics (bank-provided)

Line of business	NOP, FY26-27 (retail + group)
Life	10.33 lakh
Health	2.41 lakh
General	4.74 lakh
Total	**17.48 lakh**

Parameter	Value
Branches	1,600
Customers	1.5 crore
Specified Persons (SPs)	7,000

2. Derived platform load

Metric	Average	Peak	Assumption
Policies per day	~5,800	~17,500	300 working days; 3x peak (March / quarter-end)
Journeys per day (lead, quote, issuance)	~17,500	~52,500	3 journeys per issued policy
API calls per day	~7 lakh	~21 lakh	~40 calls per journey (CBS, KYC, insurer, payment)
Throughput (peak hour)	~40 TPS	~120 TPS sustained, ~235 TPS burst	20% of daily volume in peak hour; 2x burst
Database writes	~10 / s	~30 / s sustained, ~60 / s burst	~10 writes per journey; reads ~10x writes
Concurrent sessions	~1,200	~3,500	35% of SPs at peak, plus ~1,000 customer-side steps (OTP, consent, payment)

3. Data volumes

Store	Year 1	Year 2 (cumulative)	Provisioning
RDS: policies, leads, consents, business audit trail	~220 GB	~450 GB	500 GB gp3 with storage autoscaling to 1.5 TB
S3: proposal forms, KYC, policy documents	~1.75 TB	~3.5 TB	Lifecycle to S3-IA after 90 days; cross-region replication to DR
S3: API and application logs (via CloudWatch)	~0.4 TB	~0.8 TB	Retention as per bank policy; archive to Glacier
Redis: sessions, reference-data cache	Under 4 GB	Under 4 GB	8 GB per node

4. Infrastructure sizing

Component	Prod-DC (Mumbai)	Prod-DR (Hyderabad)	UAT (Mumbai)
Load balancer	1 x ALB, WAF attached	Provisioned on DR invocation	1 x ALB
Web tier	2 x m5a.xlarge, autoscale to 3	2 x m5a.xlarge, launched on DR invocation	1 x m5a.large
App tier	2 x m5a.xlarge, autoscale to 4	Up to 4 x m5a.xlarge, launched on DR invocation	1 x m5a.xlarge
Database (RDS MySQL)	db.m5.xlarge, Multi-AZ, 12,000 IOPS	db.m5.xlarge, cross-region read replica (always on)	db.m5.large, Single-AZ
Cache (Redis)	2 x m5a.large (primary + replica)	Launched on DR invocation	1 x m5a.large
Proxy	2 x m5a.large	Launched on DR invocation	1 x m5a.large
NAT gateway	2 (one per AZ)	Provisioned on DR invocation	1
Security and ops	KMS, Secrets Manager, CloudTrail, CloudWatch, GuardDuty, Security Hub, WAF	KMS, Secrets Manager, CloudTrail, CloudWatch, GuardDuty	KMS, Secrets Manager, CloudTrail, CloudWatch
Pricing model	1-yr Savings Plan (EC2), 1-yr RI (RDS)	1-yr RI (RDS); EC2 On-Demand when running	On-Demand, weekdays 10 hrs/day via scheduler

5. Sizing rationale

App tier: two nodes carry the average peak hour (~40 TPS) with headroom; autoscaling adds up to two nodes for March and quarter-end bursts (~235 TPS). Most request latency is in external calls (CBS, KYC, insurers), so the tier is sized on concurrency rather than raw compute.

Web tier: two nodes cover ~3,500 peak concurrent sessions; autoscaling adds a third during March and quarter-end peaks.

Database: db.m5.xlarge (4 vCPU, 16 GB) carries year-1 load (~30 writes/s sustained, ~600 reads/s burst). Multi-AZ provides automatic failover within the region. The instance is resized to db.m5.2xlarge in year 2 as policy volume grows past 35 lakh records.

DR: pilot light in a separate region (Hyderabad). The database and documents are continuously replicated (RDS cross-region replica, S3 replication); application, cache and network tiers are launched from pre-built images and infrastructure-as-code on DR invocation.

UAT: functionally identical to Prod at reduced capacity, scheduled weekdays, 10 hours a day.

6. Scaling headroom

Autoscaling on the web and app tiers covers peak periods within the stated FY26-27 volumes. Beyond that, both tiers scale horizontally, and RDS scales vertically to db.m5.2xlarge and then db.m5.4xlarge, with a read replica for MIS if required.